Secure Remote Access Management

Software Requirements Specification

IEEE 830-1998 outline

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure university remote access management system that controls off-campus access to internal university applications, servers, research systems, and administrative resources. Authorized administrators should be able to define remote-access permissions, approve or revoke access for users, review active remote sessions, and terminate suspicious sessions. The system should authenticate users before access is granted, enforce role-based access restrictions, detect repeated failed login attempts and unauthorized access attempts, generate security alerts, and maintain audit logs of remote access activity and administrator actions. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a secure university remote access management system that controls off-campus access to internal university applications, servers, research systems, and administrative resources.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- control off-campus access to internal university applications, servers, research systems, and administrative resources - FR-001
- allow Authorized administrators to define remote-access permissions - FR-002
- allow Authorized administrators to approve or revoke access for users - FR-003
- allow Authorized administrators to review active remote sessions - FR-004
- allow Authorized administrators to terminate suspicious sessions - FR-005
- authenticate users before access is granted - FR-006
- enforce role-based access restrictions - FR-007
- detect repeated failed login attempts - FR-008
- detect unauthorized access attempts - FR-009
- generate security alerts - FR-010
- maintain audit logs of remote access activity and administrator actions - FR-011
- provide a centralized dashboard - FR-012
- integrate with existing university identity services - FR-013
- support multiple campus locations - FR-014

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	users	Uses permitted functions; each source class keeps its specified permissions.
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- What conditions define repeated authentication failures?
- What conditions define unauthorized access attempts?
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?
- Which security acceptance outcomes must be demonstrated?
- What measurable reliability and recovery targets must the system meet?
- What measurable scalability targets must the system meet?

- What criteria define acceptable manageability?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-003, FR-006, FR-012. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-013. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow Authorized administrators to define remote-access permissions.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall allow Authorized administrators to approve or revoke access for users.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall allow Authorized administrators to review active remote sessions.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall allow Authorized administrators to terminate suspicious sessions.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall authenticate users before access is granted.

Traceability: E-003, TEST-006

FR-007 · Should · explicit

The system shall enforce role-based access restrictions.

Traceability: E-003, TEST-007

FR-008 · Should · explicit

The system shall detect repeated failed login attempts.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall detect unauthorized access attempts.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall generate security alerts.

Traceability: E-003, TEST-010

FR-011 · Should · explicit

The system shall maintain audit logs of remote access activity and administrator actions.

Traceability: E-003, TEST-011

FR-012 · Should · explicit

The system shall provide a centralized dashboard.

Traceability: E-004, TEST-012

FR-013 · Should · explicit

The system shall integrate with existing university identity services.

Traceability: E-004, TEST-013

FR-014 · Should · explicit

The system shall support multiple campus locations.

Traceability: E-004, TEST-014

3.3 Performance requirements

NFR-004 · Should · explicit

The system shall remain scalable.

Traceability: E-004, TEST-018

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall remain reliable.

Traceability: E-004, TEST-017

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-002, FR-006, FR-007, FR-010, FR-011.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-015

NFR-002 · Should · explicit

The system shall remain secure.

Traceability: E-004, TEST-016

3.6.4 Maintainability

NFR-005 · Should · explicit

The system shall remain manageable.

Traceability: E-004, TEST-019

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: User identities and authenticated access

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Unauthorized use of a legitimate identity could expose resources or alter system state.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-003, E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.2 THR-002: Role and permission boundaries

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A lower-privilege user could access restricted functions or sensitive resources.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.3 THR-003: Access grants and revocations

Scenario: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A formerly authorized principal could continue using access that is no longer approved.

Linked requirements: FR-002, FR-003, FR-004, FR-005

Original evidence: E-002

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.4 THR-004: Monitoring and security alerts

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Relevant suspicious activity could remain undetected or receive delayed investigation.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.5 THR-005: Source-restricted access

Scenario: A principal attempts to reach a source-restricted resource without completing the required authentication. Source boundary: The system should authenticate users before access is granted, enforce role-based access restrictions, detect repeated failed login attempts and unauthorized access attempts, generate security alerts, and maintain audit logs of remote access activity and administrator actions.

Hypothetical actor or cause: Unconfirmed actor or cause (planning hypothesis)

Potential consequence: An unauthenticated principal could gain access to the restricted resource.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

Residual exposure: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

A.6 THR-006: Integration trust paths

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Untrusted integration data could cause incorrect decisions or hide security events.

Linked requirements: FR-012, FR-013, FR-014, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.7 THR-007: Security and administrative audit evidence

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Investigators could lose reliable evidence of the events required by the brief.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

B.1 RISK-001

Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Proposed treatment: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-003, E-004

B.2 RISK-002

Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Proposed treatment: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-002, E-003

B.3 RISK-003

Derived risk affecting Access grants and revocations: A formerly authorized principal could continue using access that is no longer approved.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Proposed treatment: Trace permission changes through subsequent access decisions and any active access; confirm revocation timing and session semantics before baselining them. Assign a responsible

reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005

Original evidence: E-002

B.4 RISK-004

Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Proposed treatment: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

B.5 RISK-005

Derived risk affecting Source-restricted access: An unauthenticated principal could gain access to the restricted resource.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to reach a source-restricted resource without completing the required authentication. Source boundary: The system should authenticate users before access is granted, enforce role-based access restrictions, detect repeated failed login attempts and unauthorized access attempts, generate security alerts, and maintain audit logs of remote access activity and administrator actions.

Proposed treatment: Review the failure or abuse path against the linked source obligations. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

B.6 RISK-006

Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Proposed treatment: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-012, FR-013, FR-014, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.7 RISK-007

Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Proposed treatment: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009, FR-010, FR-011

Original evidence: E-003

C SDLC recommendation and delivery workflow

Proposed lifecycle: Incremental

The Incremental model is selected based on derived evidence showing the need for integrated security reviews (E-002, E-003), operational validation through monitoring feedback (E-003), staged deployment across multiple locations (E-004) to enable controlled rollout, and integration verification of connected systems (E-004). These factors collectively support a phased, iterative approach that aligns with security assurance, operational feedback loops, and system interoperability. The Incremental model scores highest at 66 points—outperforming Hybrid (65), DevSecOps (63), Spiral (51), and others—with derived strengths in both validation and deployment scope. While the evidence is derived rather than direct, it establishes a consistent need for iterative testing and feedback. The decision remains proposed due to unknown factors including budget, failure consequences, legacy systems, and verification methods; these are not assumed or resolved. No model demonstrates superior performance across all derived criteria, but Incremental best balances security assurance, operational validation, and deployment flexibility within the available evidence.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-002, E-003.
- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-003.
- deployment_scope = staged rollout opportunity (derived). Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. Original evidence: E-004.
- integration_assurance = integration validation (derived). Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. Original evidence: E-004.

Candidate	Suitability points
Incremental	66
Hybrid	65
DevSecOps	63
Spiral	51
Agile	40
V-Model	40
Waterfall	40

C.1.1 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002, E-003)
- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.3 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002, E-003)
- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

C.1.4 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002, E-003)
- Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.5 Agile suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.; FR-002: The system shall allow Authorized administrators to define remote-access permissions.; FR-003: The system shall allow Authorized administrators to approve or revoke access for users.; FR-004: The system shall allow Authorized administrators to review active remote sessions.; FR-005: The system shall allow Authorized administrators to terminate suspicious sessions.; FR-006: The system shall authenticate users before access is granted.; FR-007: The system shall enforce role-based access restrictions.; FR-008: The system shall detect repeated failed login attempts.; FR-009: The system shall detect unauthorized access attempts.; FR-010: The system shall generate security alerts.; FR-011: The system shall maintain audit logs of remote access activity and administrator actions.; FR-012: The system shall provide a centralized dashboard.; FR-013: The system shall integrate with existing university identity services.; FR-014: The system shall support multiple campus locations.

- Activity: Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review implementation and acceptance evidence against FR-001: The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.
- Security check: Review implementation and acceptance evidence against FR-002: The system shall allow Authorized administrators to define remote-access permissions.
- Security check: Review implementation and acceptance evidence against FR-003: The system shall allow Authorized administrators to approve or revoke access for users.
- Security check: Review implementation and acceptance evidence against FR-006: The system shall authenticate users before access is granted.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall enforce role-based access restrictions.
- Security check: Review implementation and acceptance evidence against FR-009: The system shall detect unauthorized access attempts.
- Security check: Review implementation and acceptance evidence against FR-010: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall maintain audit logs of remote access activity and administrator actions.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.2 Incremental delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.; FR-002: The system shall allow Authorized administrators to define remote-access permissions.; FR-003: The system shall allow Authorized administrators to approve or revoke access for users.;

FR-004: The system shall allow Authorized administrators to review active remote sessions.; FR-005: The system shall allow Authorized administrators to terminate suspicious sessions.; FR-006: The system shall authenticate users before access is granted.; FR-007: The system shall enforce role-based access restrictions.; FR-008: The system shall detect repeated failed login attempts.; FR-009: The system shall detect unauthorized access attempts.; FR-010: The system shall generate security alerts.; FR-011: The system shall maintain audit logs of remote access activity and administrator actions.; FR-012: The system shall provide a centralized dashboard.; FR-013: The system shall integrate with existing university identity services.; FR-014: The system shall support multiple campus locations.

- Activity: Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: In an authorized isolated test environment, execute TEST-001: Given: the system evaluates a user's off-campus access request against defined policies When: the system initiates an access control decision for a university application, server, research system, or administrative resource Expected: the system returns a response indicating whether the request is permitted or denied based on configured rules and context
- Activity: In an authorized isolated test environment, execute TEST-002: Given: An authorized administrator reviews the remote-access policy definition for a named user. When: The administrator initiates a request to define permissions for that user. Expected: The requested policy definition is returned and available for examination in the system interface.
- Activity: In an authorized isolated test environment, execute TEST-003: Given: An authorized administrator receives a request to approve or revoke access for a named user. When: The administrator approves that request through the defined workflow. Expected: The approval decision is recorded and reflected in the system state within one second of the action completion.
- Activity: In an authorized isolated test environment, execute TEST-004: Given: An authorized administrator requests a review of active remote sessions for a named user. When: The administrator submits a request to retrieve those sessions via the defined interface. Expected: The system returns the list of active sessions and their associated metadata for examination.
- Activity: In an authorized isolated test environment, execute TEST-005: Given: An authorized administrator identifies a suspicious session for a named user. When: The administrator requests termination of that session through the defined interface. Expected: The system confirms the session has been terminated and no further activity is observed from that session within five seconds of the request completion.
- Activity: In an authorized isolated test environment, execute TEST-006: Given: the system receives a principal identity from a controlled identity fixture When: the system initiates authentication for the principal Expected: authentication is accepted or rejected based on the fixture's defined credentials
- Activity: In an authorized isolated test environment, execute TEST-007: Given: the system receives a role or policy permission fixture When: the system evaluates access based on that role or policy Expected: access is allowed or denied in accordance with the fixture's defined permissions
- Activity: In an authorized isolated test environment, execute TEST-008: Given: the system observes a controlled sequence of failed login attempts When: the system processes the repeated authentication failures Expected: the condition of repeated failed attempts is identified by the system
- Activity: In an authorized isolated test environment, execute TEST-009: Given: the system receives a controlled access attempt from an unauthorized source When: the system evaluates the origin and permissions of the request Expected: the condition of unauthorized access is identified by the system

- Activity: In an authorized isolated test environment, execute TEST-010: Given: a controlled alert condition is triggered by system activity When: the system detects and responds to the event Expected: an alert exists in the designated monitoring channel
- Activity: In an authorized isolated test environment, execute TEST-011: Given: a remote access action or administrator intervention occurs When: the system records the event in its log store Expected: an entry exists in the audit log for that activity
- Activity: In an authorized isolated test environment, execute TEST-012: Given: a system-controlled user accesses the dashboard named in the source obligation When: the system opens the source-named dashboard Expected: the dashboard is displayed or available for use
- Activity: In an authorized isolated test environment, execute TEST-013: Given: a system-controlled user initiates an exchange with the named external identity service When: the system performs a controlled interaction with that service Expected: interoperation occurs as observed in the source evidence
- Activity: In an authorized isolated test environment, execute TEST-014: Given: a user initiates use from each stated campus location When: the system supports access from that location Expected: use succeeds at each location
- Activity: In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data exfiltration during remote operations. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific metrics or thresholds are provided in the current draft.
- Activity: In an authorized isolated test environment, execute TEST-016: Given: A draft acceptance plan for the goal that the system remains secure during remote access operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-017: Given: A draft acceptance plan for the goal that the system remains reliable during remote access operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-018: Given: A draft acceptance plan for the goal that the system remains scalable during remote access operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: In an authorized isolated test environment, execute TEST-019: Given: A draft acceptance plan for the goal that the system remains manageable during remote access operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.

- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review implementation and acceptance evidence against FR-001: The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.
- Security check: Review implementation and acceptance evidence against FR-002: The system shall allow Authorized administrators to define remote-access permissions.
- Security check: Review implementation and acceptance evidence against FR-003: The system shall allow Authorized administrators to approve or revoke access for users.
- Security check: Review implementation and acceptance evidence against FR-006: The system shall authenticate users before access is granted.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall enforce role-based access restrictions.
- Security check: Review implementation and acceptance evidence against FR-009: The system shall detect unauthorized access attempts.
- Security check: Review implementation and acceptance evidence against FR-010: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall maintain audit logs of remote access activity and administrator actions.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.3 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, FR-013, FR-014, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package
- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review implementation and acceptance evidence against FR-001: The system shall control off-campus access to internal university applications, servers, research systems, and administrative resources.
- Security check: Review implementation and acceptance evidence against FR-002: The system shall allow Authorized administrators to define remote-access permissions.

- Security check: Review implementation and acceptance evidence against FR-003: The system shall allow Authorized administrators to approve or revoke access for users.
- Security check: Review implementation and acceptance evidence against FR-006: The system shall authenticate users before access is granted.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall enforce role-based access restrictions.
- Security check: Review implementation and acceptance evidence against FR-009: The system shall detect unauthorized access attempts.
- Security check: Review implementation and acceptance evidence against FR-010: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against FR-011: The system shall maintain audit logs of remote access activity and administrator actions.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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